Introduction

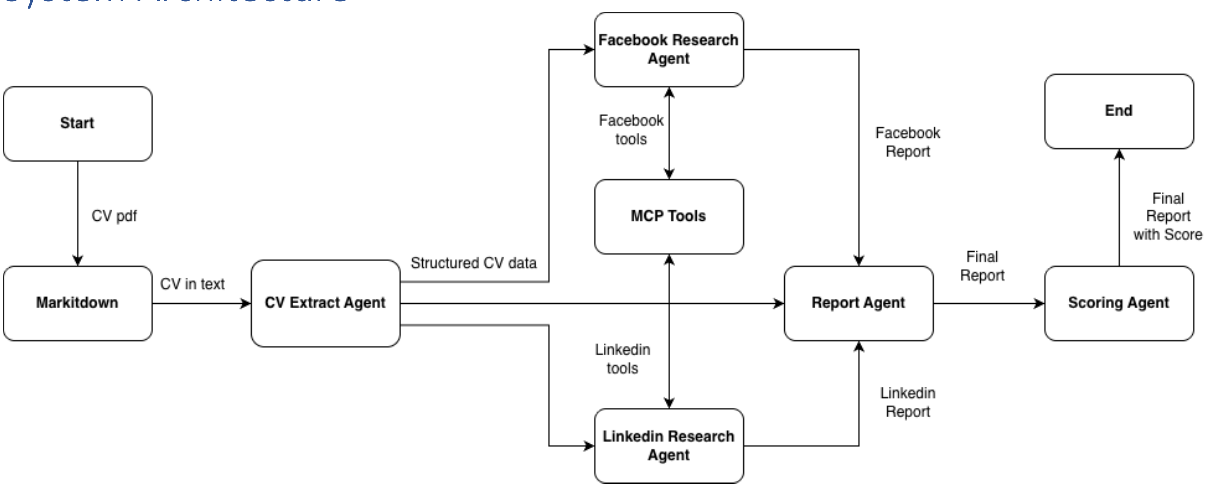
This assignment implements an AI agent system that automates CV verification against social media profiles (LinkedIn and Facebook) for Know Your Customer (KYC) and recruitment compliance processes. The system addresses the manual verification challenge by:

1. Extracting structured information from CV documents
2. Detecting discrepancies and timeline inconsistencies
3. Cross-referencing candidate claims against LinkedIn and Facebook profiles via MCP servers
4. Generating a comprehensive verification report with a trustworthiness score (0-100)

This solution uses Gemini 2.0 Flash (via Vertex AI) as the LLM for information extraction and reasoning. The system leverages the MCP server provided by instructor for LinkedIn and Facebook profile access.

This report documents the system architecture, agent workflow design, tool usage strategy, and sample verification results.

System Architecture



The solution follows a converter plus 5-agent pipeline with clear separation of concerns:

1. **Markdown Converter:** Converts PDF CV → structured markdown text
2. **CV Extract Agent:** Markdown CV -> Structured CV data to be verified
3. **Facebook Research Agent:** Structured CV + Facebook MCP tools -> Facebook Verification Report
4. **LinkedIn Research Agent:** Structured CV + LinkedIn MCP tools -> LinkedIn Verification Report
5. **Report Agent:** Struct CV + FB Report + LI Report -> Final Report
6. **Scoring Agent:** Final Report → numeric score (0-100 → 0-1)

Design Decisions

1. Hybrid LLM + Code Approach for Internal Inconsistency

Problem: LLMs miss subtle internal inconsistency, with only around 70% accuracy
Solution: this design uses LLM extracts structured data, then uses Python code validates with strict rules:

1. Timeline overlap, e.g., 2021-2024 Job A, 219 – 2025 Job B
2. Future date
3. Start date > end date
4. Multiple present job

It successfully obtained 100% overlap detection accuracy

2. Either/Or Matching Logic (Not Both Required)

Problem: Requiring both platforms penalize legitimate cases where one profile is incomplete or missing, while either one profile is enough to support the claims in CVs.
Solution: Clearly stated in the prompt - Support from either platform = valid evidence

- CV="BCG Engineer" + LinkedIn="BCG Engineer" = MATCH (Facebook irrelevant)
- Only flag when both platforms actively contradict CV

3. Strict Mismatch Hierarchy

MAJOR (LOW confidence):

- Company mismatch (EY vs PwC)
- Education mismatch (PhD vs MSc)
- Different schools

MINOR:

- Slightly different title (Senior Engineer vs Engineer)
- Unclear degree (BSc vs BSc in Engineering)
- 1 year date discrepancy

Agent Workflow and Tool Usage Strategy

Phase 1: Information Extraction

- **LLM Task:** Parse CV fields (name, education, experience, skills)
- **Code Task:** Detect overlaps using interval intersection algorithm
- **Output:** JSON with internal_inconsistent array

Phase 2: Social Media Verification

The agent performs research on the two social platforms, Facebook and LinkedIn, using MCP tools and generate intermediate reports concurrently .

Facebook Agent Workflow:

1. search_facebook_users(q="John Smith", limit=20, fuzzy=True)
2. Filter by city from output results (not input parameter)
3. get_facebook_profile(userid) until good match found
4. Apply mismatch rules (company mismatch = MAJOR, PhD vs MSc = MAJOR, Senior vs regular = MINOR)
5. Generate markdown with Match Confidence and fields breakdown

LinkedIn Agent Workflow:

1. search_linkedin_people(q="John Smith", location="Singapore") for EVERY location
    - Search city BEFORE country
    - DO NOT combine "Tokyo, Japan" - separate calls
2. Analyze headlines to pick best matches where headline/industry/location match CV title
3. get_linkedin_profile(person_id) to get profile details
4. Verify: Company Names, Job Titles, Dates, Education (University, Degree)
5. Apply mismatch rules:
    - PhD vs MSc/Master's = major + LOW/MEDIUM confidence
    - Different schools = major
    - Unclear degree (BSc vs BSc in Engineering) = minor
    - Senior vs regular title = minor
    - Different field/industry = major
    - Exact company match = strong, no match = uncertain (NOT matched)

Phase 3: Report Synthesis

Report Agent Tasks:

1. Synthesize clear verification report for human reviewer
2. Highlight overall alignment between CV and social profiles
3. Identify key matches that support candidate's claims
4. Flag key discrepancies with severity assessment
5. Note key missing education or experience in CV but missing on BOTH LinkedIn and Facebook
6. Identify areas of uncertainty (missing or ambiguous data)

Key Principle:

- SUPPORT FROM EITHER PLATFORM = MATCH (not discrepancy)
- Only include discrepancies between CV and BOTH social profiles (missing or mismatch)
- DO NOT include discrepancies between Facebook and LinkedIn profiles
- IF CV is supported by either Facebook or LinkedIn, it is in MATCH ONLY
    - CV="Engineer" + LinkedIn="Engineer" = MATCH (ignore Facebook)
    - CV="BCG" + Facebook="BCG" = MATCH (ignore LinkedIn)

Red Flags:

- Compare Company Names, Job Titles, Dates
- Compare Education (University, Degree)
- CV's experience not found on social profiles â†' major AND put in "Key Missing"
- Small date discrepancies (1 year off) = minor issues
- Major discrepancies (never worked there, wrong degree) = major red flags
- Internal consistency: check if social profile clarifies

Scoring Agent:
Produces final numeric score 0-100, and then scale down to 0-1 for evaluation, with scoring Scale:

- 100: Perfect match
- 75-99: Minor discrepancies (1yr date off, similar titles)
- 50-74: Ambiguous matches with multiple minor discrepancies or internal inconsistencies
- 35-49: Major discrepancies or major internal inconsistencies not resolved by social media
- 20-34: Many major discrepancies or major internal inconsistencies
- 0-19: Clear fabrications or no verifiable profile found

Testing Result

The system is tested towards 5 provided CVs.

1. CV_1.pdf

Name: John Smith
Report Summary: The CV of John Smith shows a good alignment with his LinkedIn profile. Key details such as his name, location, current title, education at McGill University, and experience at ByteDance as an Engineer are consistent between the CV and LinkedIn. While Facebook profiles show some matches in name and location, there are discrepancies in job title, company, and education. Overall, the LinkedIn profile supports the CV's claims, suggesting a low risk.
Final Score: 0.85 (PASS)
Ground Truth: Pass
Correctness: Yes

2. CV_2.pdf

Name: Minh Pham
Report Summary: The CV of Minh Pham shows good alignment with their LinkedIn profile, particularly regarding work experience at BCG and Tencent, as well as education at The University of Hong Kong. While there are discrepancies with the Facebook profile, the LinkedIn profile provides strong support for the CV's claims. Overall, the verification suggests a low risk profile.
Final Score: 0.85 (PASS)
Ground Truth: PASS
Correctness: Yes

3. CV_3.pdf

Name: Wei Zhang
Report Summary: The CV of Wei Zhang generally aligns well with their Facebook and LinkedIn profiles. Key details such as name, current title, company, and education are consistent across platforms. A minor discrepancy exists regarding the end year of their current role at PwC, and the CV includes location Sydney which is not mentioned in either social media profile. Overall, the available information suggests a low risk profile.
Final Score: 0.85 (PASS)
Ground Truth: PASS
Correctness: Yes

4. CV_4.pdf

Name: Rahul Sharma
Report Summary: The CV of Rahul Sharma presents several inconsistencies when compared to his online presence on Facebook and LinkedIn. While some details like name and education at Tsinghua University are confirmed, discrepancies arise concerning his current job title, company, and experience details. The overlapping dates and future dates in the experience section of the CV also raise concerns about the accuracy of the provided information.
Final Score: 0.35 (REJECTED)
Ground Truth: REJECTED
Correctness: Yes

5. CV_5.pdf

Name: Rahul Sharma
Report Summary: The verification report for Rahul Sharma reveals significant discrepancies between the provided CV and available social media profiles. While the name matches across platforms, key details regarding education, work experience, and job titles show inconsistencies. The absence of several listed companies in both LinkedIn and Facebook profiles raises concerns about the accuracy of the CV.
Final Score: 0.25 (REJECTED)
Ground Truth: REJECTED
Correctness: Yes

Accuracy

Across 5 CVs tested, the system got 5 corrects, achieving 100% accuracy.

Conclusion

In conclusion, this project delivers a multi-agent CV verification system that reliably cross-references candidate claims against LinkedIn and Facebook profiles via MCP servers. On the 5 sample CVs tested, the system achieves 100% accuracy in timeline overlap detection, discrepancy classification, and trustworthiness scoring.

GitHub Repository

The full implementation and this report are available at: https://github.com/TommyS725/FTEC5660/tree/main/homeworks/hw2